

## G. Smoothness Errors Compound With Layers

In this section, we show how Rayleigh quotient errors can be used to explain a model’s inability to fit a dataset. In particular, we show that the success of the Factorized Fourier Neural Operator can be attributed to its ability to perform (Fourier space) edge rewiring as a means of combating over smoothing.

### G.1. Models and Data

We compare Fourier neural operators developed for graphs and meshes. Fourier neural operators are a class of neural network used for PDE solving where the learnable parameters live in Fourier space in order to avoid solutions that are dependent on specific discretizations of the mesh. We compare Factorized Fourier Neural Operators (FFNOs) (Tran et al., 2023) with Geometric Fourier Neural Operators (FFNOs) (Li et al., 2023). Our choice of models is motivated by the comparisons in Tran et al. (2023) that demonstrate that the process of factorization, wherein Fourier transforms act independently on each dimension of the input, leads to improved scalability with the number of layers.

We use the structured mesh datasets from (Li et al., 2023; Tran et al., 2023). In contrast to the PyVista meshes in Sec. 6.2 which were used to stress test the expressiveness of the models, the irregular geometries in this setting are meant to be indicative of real world scenarios. Further dataset, training, and evaluation details are given in Sec. H.

### G.2. Results

As seen in Fig. 6, the Geo-FFNO model diverges at the same time it starts to over smooth. In fact, once the number of layers hit 16 for the Airfoil dataset and 20 for the Plasticity dataset, the Geo-FFNO Rayleigh quotients were within machine precision of zero. One interpretation of the result is that the factorization in FFNO rewires the mesh edge connectivity (in frequency space) and thereby prevents nodes from sending messages to their typical neighbors.

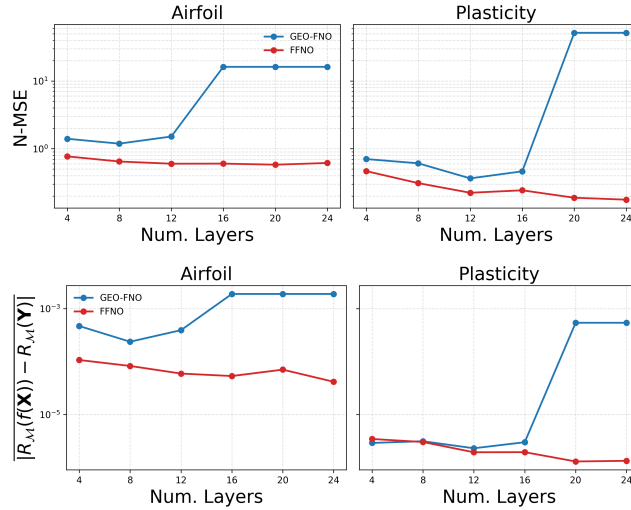


Figure 6. **Top:** NMSE as a function of the number of layers in the surrogate predictions. The FFNO model is better able to scale with the number of layers. **Bottom:** The average absolute error between the Rayleigh quotient of the ground truth and the surrogate predictions as a function of the number of layers in the surrogate. The FFNO model is better able to scale with the number of layers as it does not exhibit over smoothing.

## H. FNO Dataset, Training, and Evaluation Details

In this section we provide further experimental details for Sec. G.

### H.1. FNO Datasets

The two structure mesh datasets from (Li et al., 2023; Tran et al., 2023) are the airfoil dataset and the plasticity dataset. We recap each briefly below.

### H.1.1. AIRFOIL DATASET

The airfoil dataset considers the transonic flow over an airfoil. Let  $\rho^f$  be the fluid density,  $\mathbf{v}$  the velocity vector,  $p$  the pressure, and  $E$  the total energy. The governing equation is the Euler equation:

$$\frac{\partial \rho^f}{\partial t} + \nabla \cdot (\rho^f \mathbf{v}) = 0, \quad \frac{\partial \rho^f \mathbf{v}}{\partial t} + \nabla \cdot (\rho^f \mathbf{v} \otimes \mathbf{v} + p \mathbb{I}) = 0, \quad \frac{\partial E}{\partial t} + \nabla \cdot ((E + p) \mathbf{v}) = 0.$$

Concretely, the task is to predict a scalar field, the velocity mach number, for each input coordinate pair  $(x, y)$ .

### H.1.2. PLASTICITY DATASET

The plasticity dataset considers the plastic forging problem where a block of material  $\Omega = [0, L] \times [0, H]$  is impacted by a frictionless, rigid die at  $t = 0$ . Let  $\lambda$  be the plastic multiplier constrained by  $\lambda \geq 0$ ,  $f(\sigma) \leq 0$ ,  $\lambda \cdot f(\sigma) = 0$ , let  $C$  be the isotropic stiffness tensor with Young's modulus  $E = 200$  GPa and Poisson's ratio 0.3, and let  $\sigma_Y$  be the yield strength. The yield strength is set to  $\sigma_Y = 70$  MPa with the mass density  $\rho^s = 7850 \text{ kg} \cdot \text{m}^{-3}$ . The dynamical system is given by

$$\sigma = C : (\varepsilon - \varepsilon_p), \quad \dot{\varepsilon}_p = \lambda \nabla_\sigma f(\sigma), \quad f(\sigma) = \sqrt{\frac{3}{2}} \left| \sigma - \frac{1}{3} \text{tr}(\sigma) \cdot I \right|_F - \sigma_Y.$$

Concretely, the task is to predict the displacement vector field for all time steps given a boundary condition.

## H.2. FNO Training Details

We follow the same procedure as in [Tran et al. \(2023\)](#), using their publicly available code base <https://github.com/alasdairtran/fourierflow>. Hyper parameters are tuned for each layer number and model to achieve best performance under N-MSE loss. All runs were performed on a single H200 GPU ([NVIDIA, 2025](#)).

## H.3. FNO Evaluation Details

Following [Tran et al. \(2023\)](#), we use the normalized mean squared error

$$\text{N-MSE} = \frac{1}{B} \sum_{i=1}^B \frac{\|f(\mathbf{X}_i) - \mathbf{Y}\|_2}{\|\mathbf{Y}\|_2}$$

where  $B$  is the batch size. We observed that the errors reported in [Tran et al. \(2023\)](#) were scaled up by a factor of 100, and we do the same for consistency. Instead of using the default mesh discretizations, we use Delaunay triangulations so that we can compute the mesh Rayleigh quotient in [Definition 3](#). For the plasticity dataset, we average the Rayleigh quotient error over all data points and over all time steps.